

Rose Clock Final Project Report

Team Members

Mia Yu, Applied Mathematics and Computer Science, yanmi_yu@brown.edu

Ibrahim Shah, Computer Science, ibrahim_shah@brown.edu

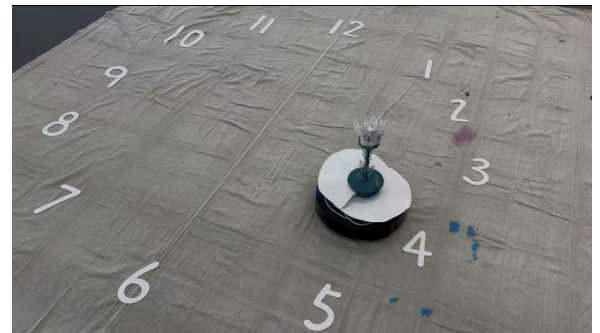
Brady Garrison, Mechanical Engineering and Computer Science, brady_garrison@brown.edu

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Project Description

Rose Clock is a robotic art installation that turns time into something the audience can physically experience. It is not a tool for controlling time, setting time, or counting down to a specific moment. Instead, it makes time feel like something that moves around us, surrounds us, and carries us forward.

The installation exists on the ground. Numbers from 1 to 12 are arranged in a circular clock-like layout. A mobile robotic system moves along this circular path, functioning as the hand of a clock. As the robot rotates, it endlessly loops around the clock face, emphasizing that time does not pause, reset, or wait for us.



Attached to the robot is a mechanical rose. Each petal is made from paper cutouts and controlled by a servo motor, allowing the petals to gradually open and close. This creates a breathing, almost lifelike motion. Instead of decaying or losing petals, the flower continuously transforms. As the project developed, we also added color to the initially white petals. This color accumulation represents the way experience builds over time. As we live, we gain memory, complexity, and emotional depth.

Technically, the system integrates Arduino-based control, servo actuation, synchronized timing logic, and a mobile ground robot into one unified loop. Conceptually, Rose Clock reframes time not as something we manage, but as something we move with.

Inspiration

Our project was inspired by the enchanted rose in *Beauty and the Beast*. In the original story, the rose marks a countdown, with falling petals showing time running out. We were drawn to this image because it turns time, an abstract force, into a visible physical transformation. We also want to represent the unprovable, irrevocable nature of time.

However, instead of recreating the rose as a countdown, we wanted to reinterpret it. Rose Clock presents time as continuous and unstoppable, not as a deadline. Time does not only disappear, instead, it also leaves traces, changes us, and adds emotional complexity.

This led to our central question: how can a robotic object make time feel like a physical presence? To explore this, we combined robotics, sculpture, embedded systems, interaction design, and installation art. A normal clock measures time, but Rose Clock tries to show what time feels like.

We chose a ground-based installation because it made the metaphor clearer. The floor becomes the clock face, the robot becomes the clock hand, and the audience experiences time as something moving through shared space.

Implementation

Our implementation had four main components: the ground robot, the clock layout, the mechanical rose, and the Arduino-based control system.

The first component was the mobile ground robot. We programmed the robot to move in a circular path around the floor. This circular movement became the core structure of the project. Instead of letting the user set a time or trigger a countdown, the robot continuously rotates around the numbers 1 through 12. This made the movement simple, readable, and conceptually clear.

The second component was the physical clock layout. We placed numbers from 1 to 12 around the robot's path so that the robot's movement could be understood as clock motion. This visual structure helps the audience immediately recognize the relationship between the robot and time.

The third component was the mechanical rose. Brady worked on the physical structure, including the initial 3D-printed prototype and updated CAD design. Mia worked on motion, material choices, and petal design. Ibrahim worked on servo motor control and actuation. The flower evolved into a multi-layered paper-cut structure, which gave the piece more visual depth while remaining feasible to build.

The fourth component was the Arduino-based control system. We used servo motors because they allowed more precise and repeatable control over the opening and closing of the petals. Arduino's Servo library is designed to control hobby servo motors and allows the shaft position to be controlled through programmed values, which fit our need for reliable petal movement.

Overall, we completed the 3D-printed prototype, updated the CAD design, developed the ground robot circular motion code, prototyped the petals, successfully programmed the servo motor code, and assembled everything together.

Changes

The project changed significantly from conception to execution, mainly to improve feasibility, reliability, and conceptual clarity.

First, we shifted from a realistic rose to an abstract paper-cut rose. A realistic flower would have required delicate fabrication and might have stayed too close to the original *Beauty and the Beast* reference. The paper-cut design gave us a cleaner visual language and made the piece feel more original.



Second, we changed the petal motion from falling to opening and closing. Falling petals would have suggested decay and countdown, which was no longer our final message. It would also have been difficult to reset for repeated demonstrations. By using servo motors, the flower became cyclical and lifelike, representing transformation rather than destruction.

Third, we switched from shape memory alloy to servo motors. Although shape memory alloy could create organic movement, it was harder to control because of heating, cooling, and recovery time. Servo motors were easier to debug, more precise, and more reliable for repeated trials.

Finally, we added color to the petals. This late-stage change emphasized the irreversible nature of time. The flower begins with white petals, but as color appears, it suggests that time leaves marks and that every experience changes us.

Completion

The project is completed. We successfully built the major components of the installation, including the robot's circular motion, the 3D-printed structure, the CAD design, the paper-cut petal prototypes, the servo motor control, and the final color treatment on the petals.

In the final version, the ground robot moves in a circular path around the clock-like floor layout, while the mechanical rose opens and closes through servo-controlled petal motion. The system brings together movement, timing, structure, and visual transformation into one complete installation.

Because this is a physical project, final testing was an important part of completion. We tested the robot's movement, the stability of the flower structure, and the reliability of the petal motion across multiple trials. These tests helped us make sure that the robot could move without destabilizing the rose and that the petals could open and close without catching or tearing.

If we had more time, we would further improve the material robustness and audience interaction of the installation. For the petals, we would add thin plastic sheets on top of the paper cutouts to make them stronger, more durable, and less likely to bend or tear during repeated motion. We would also consider adding a motion sensor so the robot or flower could respond to nearby audience members. This would make the piece more interactive while still preserving the central idea of time as something that surrounds and moves with us. We would also hide more of the mechanical structure so that the audience focuses more on the flower, motion, and time metaphor rather than the hardware.

Success

Our measure of success changed over the semester. At first, success meant building a dramatic robotic rose inspired by the enchanted rose. As the project developed, our goal became more focused: the installation would succeed if it made the audience feel time as a continuous physical presence.

By the final stage, we consider the project very successful. We met our core requirement: a ground robot moves in a circular clock-like path, while a mechanical rose opens and closes through servo-controlled petals. We also added color to the petals, which strengthened the concept by emphasizing the irreversible nature of time.

This color addition made the rose feel as if it records time, not just moves through it. The initially white petals gradually gain color, suggesting that experience



accumulates and cannot be fully undone. Time becomes not only motion, but also memory, change, and transformation.

The main possible failures were technical and conceptual. Technically, the robot needed to move smoothly and the petals needed to open and close reliably. Conceptually, the audience needed to understand the relationship between the flower and time. To make this clear, we used the visible 1 through 12 clock layout to frame the robot as the hand of a clock.

Summary

Rose Clock is a robotic installation that makes time physical. It combines a mobile ground robot, a clock-like floor layout, an Arduino-based control system, servo-actuated petals, and a transforming paper-cut rose. The project began with the enchanted rose from *Beauty and the Beast*, but evolved into a broader meditation on time as continuous, irreversible, and embodied.

The final piece turns time into something that moves around the audience. The robot endlessly circles the floor like a clock hand. The rose opens and closes like it is breathing. The petals gain color, suggesting the accumulation of experience and memory. Together, these elements invite the audience to stop thinking about time only as a tool for measurement or management, and instead feel it as a presence that surrounds and shapes us. Ultimately, Rose Clock suggests that whether we are ready or not, time keeps moving, and we move with it.

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